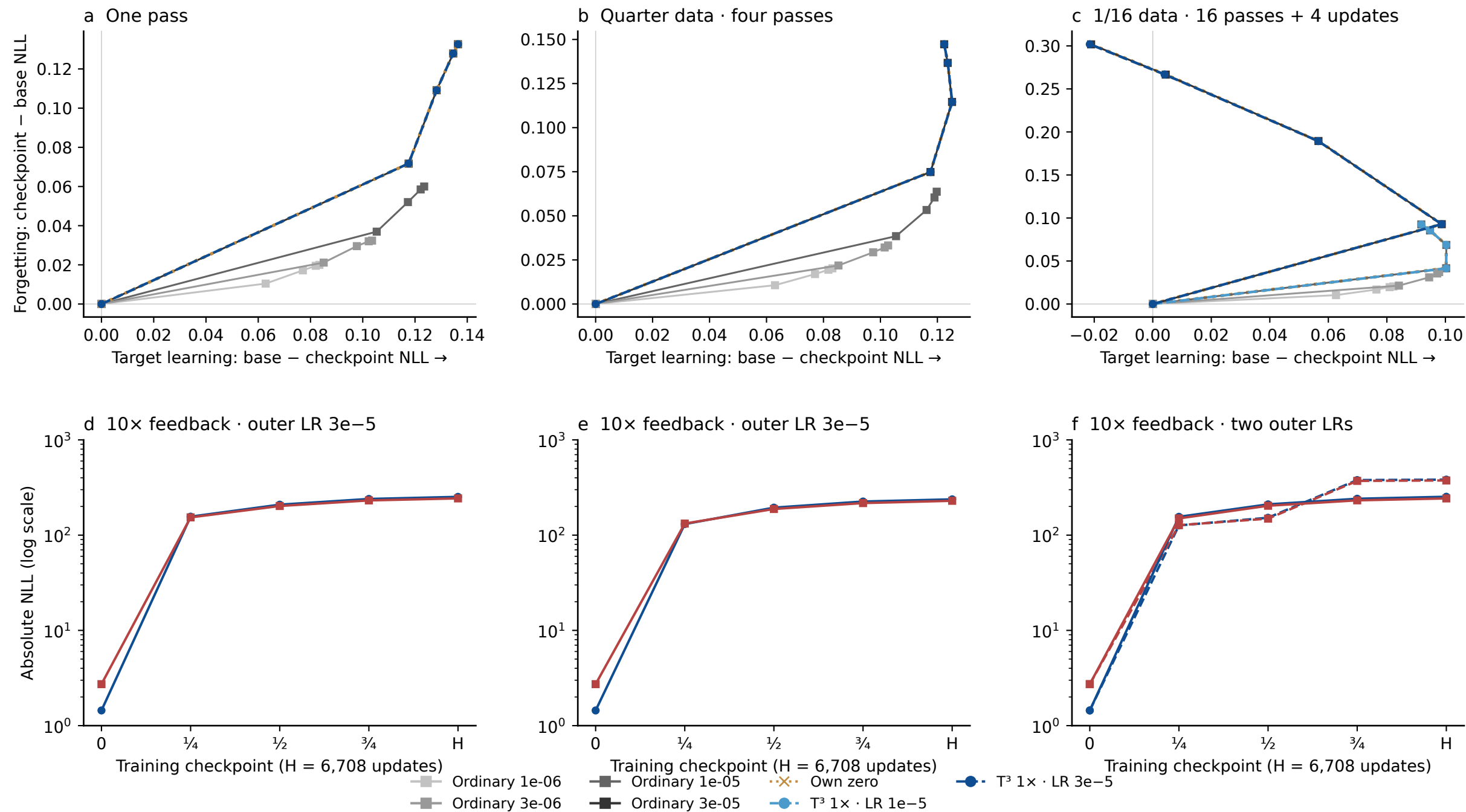


Amplifying first-order feedback does not improve CPT in this grid



Bottom row: blue = target math; red = retained FineWeb. Solid = LR 3e–5; dashed = LR 1e–5.

All 24 trajectories: 20 ordinary / zero / 1× curves above; all four 10× runs below. No results omitted.
 All 96 non-base checkpoints are measured; lines only connect fixed checkpoints. Own-zero and ordinary controls overlap.
 SmoLM2-135M, reconstructed math CPT, outer AdamW, one training seed; previously examined 512-document/domain holdouts.
 Top: more learning to the right, less forgetting downward. The three exposure regimes use different unique-data budgets.